

Deepface-Based Chaotic Image Encryption Using Key Optimization and Semi-Tensor Product Theory

Dawei Ding^{ID}, Member, IEEE, Dong Xie^{ID}, Hongwei Zhang^{ID}, Zongli Yang, and Chu'an Liu

Abstract—Currently, protecting personal privacy through selective encryption of facial images has become a research hotspot. This paper aims to design a new image encryption scheme using chaotic systems, optimization algorithms, and the Semi-Tensor Product (STP) theory. Firstly, we proposed a 3D Coupled Ikeda Map with Bounded Amplitude (3D-CIMBA), which has controllable Lyapunov exponents and high-complexity. Secondly, Particle Swarm Optimization (PSO) algorithm is used to generate control keys of the chaotic system, and produce chaotic sequences. Then, DeepFace model is applied to recognize facial regions for encryption. Moreover, the face image is encrypted by performing row-column alternation cyclic shifting operation and STP diffusion. Finally, cryptographic analysis is conducted using histograms, pixel correlation, information entropy, and SSIM. The simulation results show that this scheme demonstrates robustness against differential attacks and noise attacks, which also exhibits fast encryption speed and large key space. These results show that the proposed algorithm can encrypt facial information more efficiently and securely compared to traditional algorithms.

Index Terms—Chaotic system, image encryption, PSO optimization, DeepFace, STP theory.

I. INTRODUCTION

WITH the continuous advancement of computer technology and artificial intelligence, the information content in the Internet has experienced explosive growth in the past decade. Massive data, including images, videos, documents, and chat records, need to be transmitted through public channels, making personal privacy leakage a major challenge faced by everyone [1]. Therefore, secure communication has become a hot research topic in modern society [2]. Digital image, characterized by high redundancy, big data capacity, and strong correlation between information [3], [4], imposes more stringent requirements on encryption schemes. Many scholars are dedicated to finding encryption schemes which are more efficient and less vulnerable to attacks than traditional solutions [5]. Lots of techniques, such as DNA encoding [6], [7], compressive sensing [8], [9], [10] and information hiding [11], [12], [13], have been developed and are widely combined with image encryption.

Received 11 October 2024; revised 24 December 2024; accepted 2 February 2025. Date of publication 19 February 2025; date of current version 4 July 2025. This article was recommended by Associate Editor W. Wang. (Corresponding author: Hongwei Zhang.)

The authors are with the School of Electronic Information Engineering, Anhui University, Hefei 230601, China (e-mail: dwding@ahu.edu.cn; 695885681@qq.com; hwzhang@ahu.edu.cn; 07115@ahu.edu.cn; 166118780@qq.com).

Digital Object Identifier 10.1109/TCSVT.2025.3543824

Since the concept of chaos was first proposed in 1963, chaos theory has developed rapidly, gradually evolving into a mature and practical theoretical framework [14]. Chaos systems exhibit unpredictability and sensitivity to initial conditions [15], which means that the long-term iterative results of chaotic systems are unpredictable, and small changes in the initial state can lead to drastic changes in their dynamic behavior [16], [17]. These characteristics makes these systems widely used in the field of secure communication, especially in image encryption. According to the definition of chaotic system, it can be divided into two types: discrete systems and continuous systems. Continuous systems, such as the Chen system and Lorenz system, take longer for iterative solving, whereas discrete systems have an advantage in terms of computational cost. In recent years, many encryption schemes based on discrete chaotic systems have been proposed [18]. Zhu et al. [19] proposed a 2D Logistic-modulated-Sine-Coupling-Logistic chaotic map (2D-LSMCL) to generate chaotic matrices in image encryption. Qiu et al. [20] used a hyperchaotic map 2D-CSCM to design an encryption algorithm based on Rubik's Cube scrambling. Kocak et al. [21] proposed a new PSO key optimization algorithm based on the Modular Integrated Logistic Exponential (MILE) map.

However, many existing chaotic image encryption schemes have some short-comings [22]. For example, some of the chaotic systems are insufficient in chaotic properties, having simple structures and small range of system parameters, which may lead to chaos degradation [23]. Ding et al. [24] introduced the concept of controllable Lyapunov exponents, and by using a modulo function, they ensured that the system parameters exhibit chaotic behavior over a wide range. In addition, traditional encryption methods consist of two steps: scrambling and diffusion [25]. The XOR pixel diffusion scheme is widely used, making it susceptible to targeted attacks. Therefore, many researchers are seeking new pixel confusion schemes that differ from traditional methods, such as STP diffusion. Gao et al. [26] used STP theory and a new map called 2D-LAIC to encrypt the 3D model. Gao et al. [27] used Histogram of Oriented Gradients (HOG) to detect human face and introduced STP diffusion in facial area encryption.

Most conventional encryption schemes with good results have high time complexity. The Region of Interest (ROI) encryption compensates for the shortcomings of the aforementioned time cost [28]. Shen et al. [29] proposed an optical selective encryption scheme which was adapted in the ROI of medical image. Jamel et al. [30] introduced ROI encryption

methods into medical image area, which aimed to reduce the data size of telemedicine.

Facial images carry unique biometric information, making facial encryption an important area in ROI (Region of Interest) encryption. In applications such as face recognition and face unlocking, facial images often contain significant redundancies. Honsny et al. [31] used YOLOv3 to detect human face in video and applied fast block scrambling technique to encrypt it. Du et al. [32] proposed a new chaotic system called NCMLP and applied it in a novel multiple face image encryption scheme. The aforementioned facial encryption schemes show improvements in encryption time overhead compared to traditional methods. However, there is still room for improvement in the models and accuracy of facial detection and recognition.

Based on above analysis, a novel selective color image encryption scheme is proposed, which utilizes techniques such as DeepFace model, STP diffusion, and PSO secret key optimization. In summary main contributions of this paper are listed as follows:

1) A novel 3-dimensional chaotic map 3D-CIMBA is introduced, with controllable Lyapunov exponents, which has a wide chaos range and high complexity.

2) A face recognition encryption scheme based on the DeepFace model is employed, in order to effectively enhance encryption speed.

3) The PSO optimization algorithm is utilized for generating initial values and control parameters in the system. That produces keys associated with the image to be encrypted, which also makes the encryption scheme effective in resisting known plaintext attacks.

4) STP, as a novel scheme for pixel confusion in images, differs significantly from conventional XOR-based confusion methods, making it effective against conventional attack methods.

The rest of the paper is organized as follows:

Section II discusses the relevant technologies used in the encryption scheme of this paper. The proposed chaotic system and its dynamic behavior analysis are introduced in Section III. An image encryption and decryption scheme based on the aforementioned theories is presented in section IV. Section V conducts a cryptographic analysis of encryption and decryption. The last part provides a summary of this paper.

II. PREPARATION

A. DeepFace

DeepFace is a cutting-edge neural network architecture developed by Facebook AI Research (FAIR) for facial recognition tasks [33]. In application, DeepFace has demonstrated remarkable performance, achieving an accuracy rate of 97.35% on the standard Labeled Faces in the Wild (LFW) benchmark [34]. This level of accuracy is comparable to, and in some cases exceeds, human performance, marking a significant milestone in the quest for reliable automated facial recognition technology [35].

Utilizing the DeepFace neural network model, we can easily perform facial region recognition and comparative analysis. Its functionalities include but are not limited to

identifying gender, race, age, facial expressions, and other features of individuals. It also offers face comparison capabilities and provides a confidence score for the comparison results for reference. With this, we can establish our own facial database and selectively encrypt the faces of selected individuals.

B. PSO Algorithm

Particle Swarm Optimization (PSO) is a population-based metaheuristic algorithm that mimics the social behavior of bird flocking.

The basic idea of the PSO algorithm is to randomly initialize a swarm of particles in the search space, with each particle representing a point in the search space. Each particle has two attributes: position and velocity. The position represents a potential solution, while the velocity determines the particle's movement direction and distance [36]. The particles update their positions by tracking two 'best' solutions: one is the best position the particle has experienced (individual best solution, $pbest$), and the other is the best position experienced by the entire swarm (global best solution, $gbest$). During the execution of the algorithm, all particles in each generation update their velocities and positions according to the following formula [37]:

$$\begin{cases} v_i^{t+1} = wv_i^t + c_1r_1(pbest_i - x_i^t) + c_2r_2(gbest - x_i^t) \\ x_i^{t+1} = x_i^t + v_i^{t+1}, \end{cases} \quad (1)$$

here, v_i^t and x_i^t represent the velocity and position of the i th particle at iteration t , respectively. w is the inertia weight, controlling the preservation of particle velocity; c_1 and c_2 are learning factors determining the degree to which particles are attracted towards their personal and the global best positions; r_1 and r_2 are random numbers within $[0,1]$, introduced to add stochasticity to the search.

C. STP

There are two matrixes $P \in M_{m \times n}$, $Q \in M_{p \times q}$. If the least common multiple of n and p is t . T is defined as the semi-tensor product of the two:

$$T = P \propto Q = (P \otimes I_{t/n})(Q \otimes I_{t/p}), \quad (2)$$

where M represents the set of all $m \times n$ -sized matrices, $\propto$ denotes the semi-tensor product, $\otimes$ denotes the tensor product of matrices, and I represents the identity matrix.

Example 1: There are two matrices, $P = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix}$, and $Q = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$.

$$\begin{aligned} T = P \propto Q &= (P \otimes I_{4/4})(Q \otimes I_{4/2}) = (P \otimes I_1) \cdot (Q \otimes I_2) \\ &= \begin{bmatrix} 1 & 0 & 2 & 1 \\ 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} \cdot \left(\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) \end{aligned}$$

$$\begin{aligned}
&= \begin{bmatrix} 1 & 0 & 2 & 1 \\ 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 2 & 0 & 1 & 0 \\ 0 & 1 & 0 & 2 & 0 & 1 \\ 0 & 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 & 0 & 2 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & 4 & 1 & 5 & 2 \\ 2 & 1 & 4 & 2 & 2 & 1 \\ 0 & 1 & 1 & 3 & 2 & 3 \end{bmatrix}. \quad (3)
\end{aligned}$$

Reversely, when given matrices T and P , Q can be calculated by

$$\begin{aligned}
Q \otimes I_2 = T \cdot P^{-1} &= \begin{bmatrix} 1 & 0 & 4 & 1 & 5 & 2 \\ 2 & 1 & 4 & 2 & 2 & 1 \\ 0 & 1 & 1 & 3 & 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 2 & 1 \\ 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix}^{-1} \\
&= \begin{bmatrix} 1 & 0 & 2 & 0 & 1 & 0 \\ 0 & 1 & 0 & 2 & 0 & 1 \\ 0 & 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 & 0 & 2 \end{bmatrix} = \begin{bmatrix} I & 2I & I \\ 0 & I & 2I \end{bmatrix}. \quad (4)
\end{aligned}$$

Then, we can easily restore Q from the equation. It is worth noting that since P is not a square matrix, the operation P^{-1} corresponds to finding its generalized inverse matrix. In practice, in our encryption steps, we usually use the doubled dimension semi-tensor product to simplify our calculations. That is, n is an integer multiple of p .

III. PROPOSED CHAOTIC MAP

In this section, a new type of maps based on Ikeda is introduced. We extensively investigated the dynamical behavior of this system under various system parameter ranges and compared its performance with other mappings.

A. Ikeda Map and Proposed Map

The Ikeda map is a two-dimensional discrete chaotic system originally proposed by Kunihiko Ikeda in 1979 to model the complex dynamics of light transmission through a nonlinear optical medium. It is defined by the following complex mapping:

$$U_{n+1} = a + bU_n e^{i(\delta - \frac{K}{1+|U_n|^2})}. \quad (5)$$

In its more common real form, the Ikeda map can be expressed by (6).

$$\begin{cases} \phi_n = \delta - \frac{K}{1+x_n^2+y_n^2}, \\ x_{n+1} = a + b(x_n \cos(\phi_n) - y_n \sin(\phi_n)), \\ y_{n+1} = b(x_n \sin(\phi_n) + y_n \cos(\phi_n)), \end{cases} \quad (6)$$

where a , b , δ , K are system parameters, the map exhibits chaotic behavior when $b > 0.6$ and $x_1, y_1 \in (0, 1)$ [23]. However, the chaotic system exhibits characteristics of having a small chaotic range of parameters and poor performance in dynamical behavior.

The small chaotic range and Lyapunov exponent of the Ikeda map pose a risk that the chaotic sequences it generates through iterations may be cracked. Therefore, an improved 3-dimensional coupled Ikeda map with bounded amplitude (3D-CIMBA) is proposed. By introducing a third dimension through modifications to the mathematical expression and adding a linear term $e^g x_n$ to the original equation.

So that, by using a clipping function to restrict the function's value between 0 and 1, we obtained the final system equation. It effectively enhances the system's chaotic characteristics and the chaotic range, making it more unpredictable. The system formula of 3D-CIMBA is given as (7) and (8).

$$\phi_n = \delta - \frac{K}{1+x_n^2+y_n^2+z_n^2}, \quad (7)$$

$$\begin{cases} x_{n+1} = \text{mod}(a + b(x_n \cos(\phi_n) - y_n \sin(\phi_n)) + e^g x_n, 1), \\ y_{n+1} = \text{mod}(b(x_n \sin(\phi_n) - y_n \cos(\phi_n)) + e^g y_n, 1), \\ z_{n+1} = \text{mod}(b(x_n \sin(\phi_n) - y_n \sin(\phi_n)) + e^g z_n, 1), \end{cases} \quad (8)$$

where ϕ is an Intermediate variable while x_n , y_n and z_n represents the result of the n th iteration of three different chaotic sequences. a , b , δ , K , g are five control parameters of 3D-CIMBA, $\text{mod}(\dots, 1)$ represents the modulo operation. Through the improved system, we can generate the pseudo-random sequences required for encryption through multiple iterations.

B. Controllable Lyapunov Exponent

Assuming there is an autonomous system $\frac{dx}{dt} = F(x)$, we have the following linear differential equation:

$$\begin{cases} \frac{d\Delta x(t)}{dt} = \mathbf{J}\Delta x(t) \\ LE_i = \lim_{t \rightarrow \infty} \frac{1}{t} \ln \lambda_i(\mathbf{J}), \end{cases} \quad (9)$$

where $\mathbf{J}$ is the Jacobian matrix of the system, $i = 1, 2, 3, \dots$ [38].

Lyapunov exponent (LE) is an important index to evaluate the chaotic behavior of system motion. A positive LE can be a reference of judgment that a dynamical system is chaotic, which indicates that the trajectory of the system diverges over time [39]. For multidimensional systems, multiple positive LE indicate that their trajectories diverge in multidimensional space as well.

We can represent the Jacobian matrix of 3D-CIMBA using the following formula:

$$\mathbf{J} = \frac{\partial \mathbf{F}}{\partial \mathbf{x}} = \begin{bmatrix} j_{11} + e^g & j_{12} & j_{13} \\ j_{21} & j_{22} + e^g & j_{23} \\ j_{31} & j_{32} & j_{33} + e^g \end{bmatrix} \quad (10)$$

Take j_{11} as an example,

$$\begin{aligned} j_{11} &\leq |b|(|\cos \phi_n| + |\frac{2Kx_n(x_n \sin \phi_n + y_n \cos \phi_n)}{(1+x_n^2+y_n^2+z_n^2)}|) \leq |b||1+4K| \\ &\quad (11) \end{aligned}$$

Similarly, it can be proved that j_{mn} ($m = 1, 2, 3; n = 1, 2, 3$) are all less than or equal to $|b||1+4K|$. So that, when the parameters satisfy that $|b||1+4K| \ll e^g$, We can approximate that $j_{mn}/e^g = 0$. And the Jacobian matrix $\mathbf{J}$ can be rewrite as

$$\mathbf{J} = e^g \begin{bmatrix} \frac{j_{11}}{e^g} + 1 & \frac{j_{12}}{e^g} & \frac{j_{13}}{e^g} \\ \frac{j_{21}}{e^g} & \frac{j_{22}}{e^g} + 1 & \frac{j_{23}}{e^g} \\ \frac{j_{31}}{e^g} & \frac{j_{32}}{e^g} & \frac{j_{33}}{e^g} + 1 \end{bmatrix} = \begin{bmatrix} e^g & 0 & 0 \\ 0 & e^g & 0 \\ 0 & 0 & e^g \end{bmatrix} \quad (12)$$

Obviously, $\mathbf{J}$ has three positive identical eigenvalues e^g .

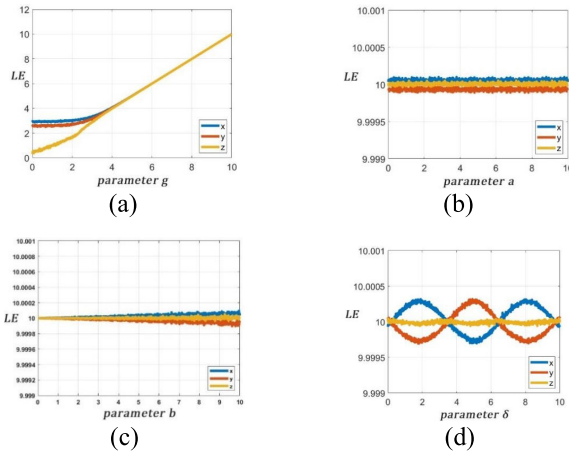


Fig. 1. LEs of 3D-CIMBA's 5 parameters. (a) parameter g ; (b) parameter a ; (c) parameter b ; (d) parameter δ .

Therefore, $\mathbf{J}$ has three identical eigenvalues. Thus, based on the LE calculation formula, the LE of 3D-CIMBA equal to g . Within the above value ranges, 3D-CIMBA consistently exhibits hyperchaotic behavior [40].

Next, the proposition above is verified through simulations. We set the initial value $(x_1, y_1, z_1) = (0.1, 0.2, 0.3)$, and the parameters $(a, b, \delta, K) = (10, 10, 0.4, 6)$. By adjusting the value of g , we can control the overall level of the LE of the system. As shown in Fig. 1(a), as the value of g increases, the Lyapunov Exponent spectrum (LEs) of the three sequences tend to be consistent and numerically almost equal to g . With g fixed at 10 and other values unchanged, the curves of LE of the parameters a, b, δ are shown in Fig. 1(b-d). Any of these parameters do not significantly affect the LEs. The above analysis indicates that for different encryption schemes, we can flexibly select the corresponding system parameters, especially g , to adjust the complexity of the system.

C. Bifurcation Diagram

A bifurcation diagram is a visual representation used in the study of dynamical systems, including chaotic systems, to illustrate how the behavior of the system changes as a parameter within the system is varied. These diagrams show how a system transitions from simple, predictable behavior to complex, chaotic dynamics through a series of bifurcations. A bifurcation occurs when a small change in the parameter value causes a sudden qualitative change in the behavior of the system. Fig. 2(a)-(b) demonstrates the bifurcation diagram of classic Ikeda map's 2 different parameters, while Fig. 2(c)-(d) shows that of 3D-CIMBA. Similarly, we compared other classic three-dimensional mapping models, such as VL-DMSIN [40]. We plotted the bifurcation diagrams for parameters b_0 and b_2 separately, as shown in Fig. 2(e)-(h). It can be observed from the figures that the proposed system has a much bigger chaotic parameters range than the Ikeda map.

D. Trajectory Diagram

The trajectory diagram of a chaotic system is a critical tool for describing the dynamic behavior of the system. A phase

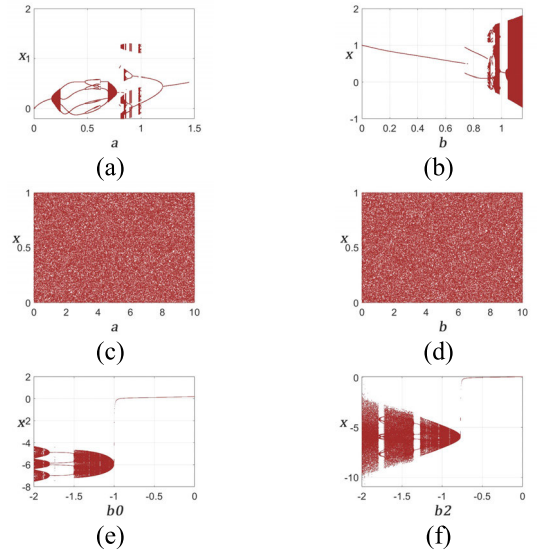


Fig. 2. Bifurcation diagram. (a)-(b) parameter a and b from Ikeda map; (c)-(d) parameter a and b from 3D-CIMBA; (e)-(f) parameter b_0 and b_2 from VL-DMSIN.

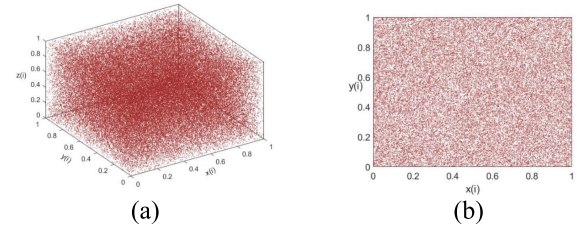


Fig. 3. Trajectory diagram of 3D-CIMBA. (a) 3D graph; (b) x-y plane projection diagram.

diagram graphically displays the path of system states over time, providing an intuitive perspective for understanding the system's long-term behavior. If the system exhibits closed-loop trajectories, this usually indicates periodic behavior. We set the system iteration count at 50,000, discarding the first 1,000 points. Fig. 3 shows the three-dimensional phase space distribution of the 3D-CIMBA system and its projection on the x-y planes. The overall distribution of the system's iteration results in the phase space is uniform, which indirectly confirms the system's chaotic nature.

E. Sample Entropy

From the previous analysis, it has been proved that when the parameter g fixed, the changes in b and k have the greatest impact on the system. Fig. 4(a) shows the results of the Sample Entropy (SE) as the parameters vary. Sample Entropy (SampEn) is an improved version of Approximate Entropy (ApEn) and is a statistical measure used to assess the complexity of time series data. It primarily evaluates the irregularity and complexity of time series. Fig. 4(b-f) shows the comparison of SE values between 3D-CIMBA and other systems.

IV. IMAGE ENCRYPTION AND DECRYPTION ALGORITHM

The overall encryption process is illustrated in Fig. 5, which can be divided into three main steps. Additionally,

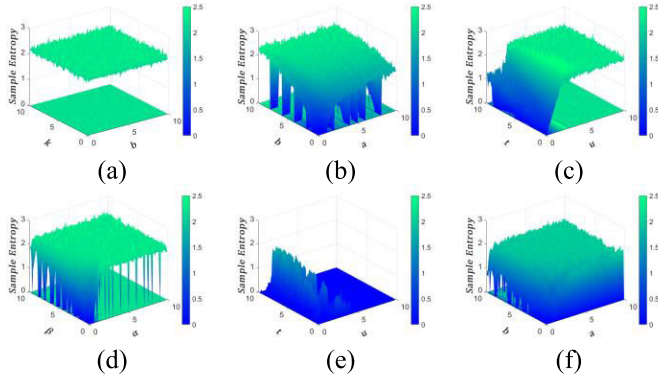


Fig. 4. SE of different chaotic system. (a) 3D-CIMBA; (b) 2-D cubic map; (c) MILE; (d) 2D-LTMM; (e) 2D-LSMCL; (f) 2D-CSCM.

TABLE I
DIFFERENCE BETWEEN DIFFERENT ALGORITHM

Algorithm	ROI encryption	STP diffusion	Chaotic system	Key optimization
Ref. [20]	×	×	2D	×
Ref. [21]	×	×	1D	✓
Ref. [29]	✓	×	2D	×
Ref. [27]	✓	✓	2D	×
Ours	✓	✓	3D	✓

in Table I, we compare the similarities and differences between the encryption algorithm used in this paper and some existing algorithms.

A. Key Optimization Based on PSO Algorithm

First, 100 particles are randomly initialized for optimization search, where the coordinates of each particle represent the initial parameters of 3D-CIMBA. Then chaotic sequences are generated to encrypt a portion of size 64*64 which is randomly cropped from the input image. Therefore, we can calculate the fitness function for PSO by (13)

$$F(p) = \omega_1 \sum_{i=1}^3 PSNR(P_i) + \omega_2(8 - H(P)) \quad (13)$$

where $PSNR(P_i)$ means Peak Signal-to-Noise Ratio of the 64*64-pixel block's RGB channels and $H(P)$ is the information entropy of that. ω_1 and ω_2 are two weight coefficients of it. PSNR is a commonly used metric to evaluate the quality of reconstructed or compressed images or videos. It measures the ratio between the maximum possible power of a signal (the "peak signal") and the power of corrupting noise that affects the fidelity of the signal (the "noise"). PSNR is expressed in decibels (dB) and is calculated using the mean squared error (MSE) between the original and the reconstructed images [41].

$$\begin{cases} MSE = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x}_i)^2 \\ PSNR = 10 \cdot \log_{10}(\frac{255^2}{MSE}), \end{cases} \quad (14)$$

Here, x_i represents the i th pixel of the image, and $\bar{x}_i$ represents the average pixel value of the image. The formula for

Algorithm 1 Secret Keys Optimization

Input: Image img , Population number $Nnum$, Dimension dim , Iterations $iter$

1. Initialize position limits x_limit and velocity limits v_limit
2. **For** each dimension i up to dim :
3. Initialize position x randomly within x_limit
4. Initialize velocity v randomly within v_limit
5. Initialize p_best as x
6. Initialize f_best as zeros
7. Initialize fp_best (individual's historical best fitness) as negative infinity
8. Initialize fg_best (global best fitness) as negative infinity
9. Initialize $w = 0.8, c_1 = 0.5, c_2 = 0.5$
10. Initialize a recorder $record$ to zero
11. Iteration starts:
12. **For** $i = 1 : iter$:
13. Calculate current fitness $F(p)$ by (13)
14. **For** $j = 1 : Nnum$
15. **If** $fp_best(j) < F(j)$:
16. Update $fp_best(j)$ and position
17. **If** $fg_best(j) < \max(fp_best)$:
18. Update $fg_best(j)$ and position
19. Update v by Eq.(1), adjust v within v_limit
20. Update x by Eq.(1), adjust x within x_limit
21. Record the global best fitness to recorder
22. Calculate the average position for all dimensions

Output: p_best (secret keys)

calculating information entropy is shown in (15)

$$H(x) = \sum_{i=1}^{256} p(x_i) \log_2 \frac{1}{p(x_i)} \quad (15)$$

For a randomly distributed irregular image, its theoretical entropy value is 8. In practical encryption, the closer the entropy of the encrypted image is to 8, the better the encryption effect.

Algorithm 1 provides pseudo-code for the specific PSO optimization process of the key. The meanings of various parameters have been given in the previous text and are not repeated here.

B. Face Recognition and Database Matching

Our facial test set comes from the Internet Movie Database (IMDB) [27]. Due to the large overall volume of the files, 3,000 images from IMDB.0 is used, with a total size of 1.3GB, to construct the facial database. Additionally, 5 images not in the database is selected for recognition testing, with the test results presented in Section VI.

In DeepFace model, whether two images belong to the same person is determined by comparing the Euclidean distance between them and a preset threshold.

If the distance between two facial feature vectors is less than this threshold, two images show the same person; otherwise, they are different ones. This threshold is usually determined by conducting experiments on a certain dataset and based on the accuracy and recall rate of face verification. Different datasets and application scenarios may require different threshold settings. In our simulation we set threshold of Euclidean distance at 0.5. If the comparison is successful, the image would be encrypted.

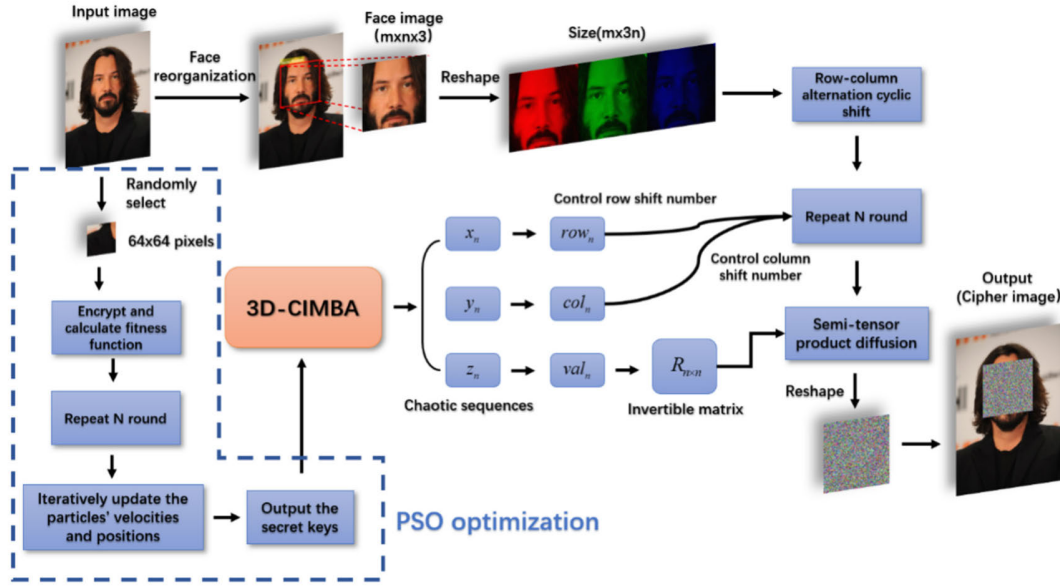


Fig. 5. Flow chart of proposed encryption method.

C. Encryption Scheme

Step1: Read the original image P , and extract the face image F of size $m \times n \times 3$ based on the previous face recognition results. Then we reshape F into the size of $m \times 3n$.

Step2: Utilizing the algorithm mentioned in Section V-A, search for the optimal key of 3D-CIMBA for the plaintext image. Here, considering the computational cost of multiple optimization rounds and the marginal effect of the optimization algorithm on improving the chaotic performance, we set the dimension of the particle swarm $\dim = 8$ (corresponding to 5 parameters and 3 initial values), the population size $N_{num} = 100$, the number of iterations $iter = 100$, the coefficient of the fitness function $\omega_1 = 0.3, \omega_2 = 0.7$.

Step3: Based on the aforementioned keys, generate the corresponding three chaotic sequences x, y, z . The first 500 states of the sequence were discarded and obtain $line, row$ and val using the operations in (16).

$$\begin{cases} line_i = \text{floor}(\text{mod}(x_i * 10^5, 3n)) \\ row_i = \text{floor}(\text{mod}(y_i * 10^5, m)) \\ val_i = 2z_i. \end{cases} \quad (16)$$

Step4: Perform row-column alternation cyclic shifting operation on the face image using sequences $line$ and row , as depicted in the algorithm 2. In the following encryption, we set the number of Iterations as 6000.

Step5: Firstly, the first n^2 elements of sequence val is rearranged to form a square matrix $R_{n \times n}$. Then, we calculate the semi-tensor product of $R_{n \times n}$ and F_2 , denoted as F_3 .

$$cipher = \text{mod}(F_2 \otimes R, 2^8), \quad (17)$$

During the decryption process, we need to compute the inverse matrix of $R_{n \times n}$, hence it is essential to ensure that matrix is invertible. If the first generated $R_{n \times n}$ is not invertible, then one should start taking values from the 101st element of val to regenerate it. If it is still not invertible, continue to take

Algorithm 2

Input: face image F , Number of Iterations $roundnum$

Output: cipher face image $C1$

1. For $i = 1 : roundnum$
2. Row Index $ix = \text{mod}(i, m)$
3. Column Index $iy = \text{mod}(i, n * 3)$
4. $F2(ix+1,:) = \text{circshift}(F1(ix+1,:), line(i));$
5. $F2(:,iy+1) = \text{circshift}(F1(:,iy+1), row(i));$
6. $C1 = \text{reshape}(F2, \text{size}(F))$

values from every subsequent 100 elements until an invertible full-rank matrix is produced. At this point, we have obtained the final encrypted result, *cipher*.

Furthermore, as the semi-tensor product operation involves a mod function, constituting an asymmetric encryption, we also need to calculate the private key key_s required for decryption [27].

$$key_s = \text{floor}(F_2 \otimes R / 2^8) \quad (18)$$

Since the encryption scheme we proposed is a type of symmetric encryption, the decryption process essentially involves operations that are the reverse of the encryption process.

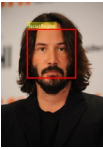
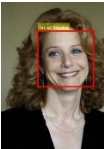
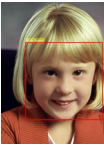
V. CRYPTANALYSIS OF PROPOSED SCHEME

In this Section, simulation tests would be conducted on the proposed recognition encryption system. The hardware configuration used is as follows: the CPU is an Intel Core I5 13600KF, with memory being DDR4 4000 16GB*2. The simulation software includes Spyder IDE 5.4.3 (Python 3.11) and MATLAB R2022a.

A. Simulation Results

We firstly tested the facial database comparison feature, using the following five images (rm342264064_1964-9-2_

TABLE II
FACE RECOGNITION AND MATCHING RESULTS

images	image 1	image 2	image 3	image 4	image 5
Input image					
Matched image					None
Results	True	True	True	True	False
Distance	0.220089	0.292698	0.255430	0.248038	\
	4802466	6778560	6057249	4743941	
	128	474	438	091	
Vertex coordinates	'x': 73, 'y': 78	'x': 73, 'y': 78	'x': 67, 'y': 111	'x': 61, 'y': 85	\
Image dimensions	'w': 133, 'h': 133,	'w': 140, 'h': 140,	'w': 120, 'h': 120,	'w': 125, 'h': 125,	\

2009.jpg, rm304650752_1955-5-16_2001.jpg, rm3388605-44_1961-7-30_1999.jpg, rm3794573056_1985-7-18_2007.jpg, rm848858368_1978-9-23_2010.jpg) as input. For convenience, these will be abbreviated as 'image 1 to 5'. Four of them were successfully matched to the image in the database with the smallest Euclidean distance, while one failed, as shown in the Table II.

Furthermore, we performed encryption and decryption operations on the facial parts of these five images that passed the test. The results are shown in the is the original image with a successfully matched facial area, which is encrypted using the process mentioned in Section IV to produce Fig. 6.

In addition, to facilitate comparison with other encryption schemes not using images from the IMDB dataset, we also performed both global and facial area-specific encryption on some commonly used images in the standard image test set (baboon, peppers, 4.1.01 and 4.1.04). To verify the proposed scheme's ability to resist plaintext attacks, we selected two $512 \times 512 \times 3$ all-black and all-white images to test the encryption effect. The encrypted results show no significant difference compared to other colored images. The images used and the results of encryption are shown in the Fig. 7.

B. Histogram and Entropy Analysis

The histogram is a graphical representation of the distribution of pixel values in an image. Typically, the pixel values in the original image exhibit a non-uniform distribution [42]. Fig. 8 displays the histogram distributions of the three RGB channels of the image before and after global encryption. From the figure, it can be observed that the pixel value distribution of the encrypted face image tends to be uniform, which does not intuitively reveal useful information.

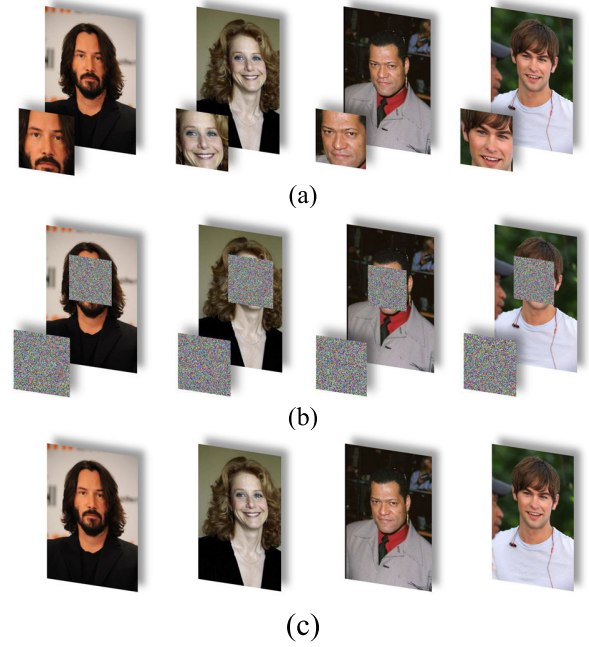


Fig. 6. Encrypted and decrypted results of facial image. (a) plain image, (b) cipher image, (c) decrypted results.

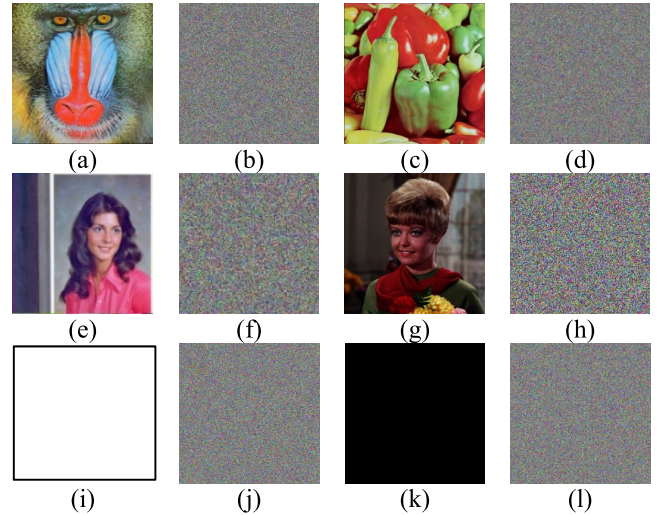


Fig. 7. Encryption and decryption of (a)-(b) baboon, (c)-(d) peppers, (e)-(f) 4.1.01, (g)-(h) 4.1.04, (i)-(j) all-white image, (k)-(l) all-black image.

The formula for calculating information entropy has been provided earlier, and the information entropy calculated according to (15) is presented in Table III.

C. Correlation Analysis

In the application of image encryption, the ideal encryption algorithm should minimize the correlation between adjacent pixels in the image to the greatest extent, making the encrypted image appear as similar to random noise as possible, thereby enhancing the security of the image. Mathematically, the correlation between adjacent pixels can be quantified by calculating the correlation coefficient between pixel values. For a given image, one may choose horizontally adjacent, vertically adjacent, or diagonally adjacent pixel pairs for this

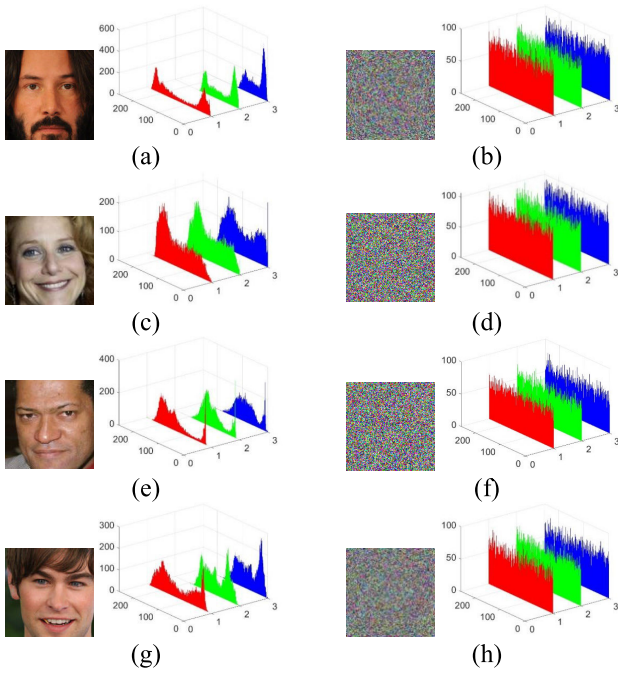


Fig. 8. Histogram of face images:(a), (c), (e), (g) plain image;(b), (d), (f), (h) cipher image.

TABLE III
ENTROPY OF TESTED IMAGES

Image	Image size	Original images			Encrypted images		
		R	G	B	Red	Gree	Blue
Face1	133×133	7.531	6.979	6.584	7.988	7.989	7.988
Face2	140×140	7.637	7.622	7.601	7.988	7.989	7.989
Face3	120×120	7.273	7.086	7.179	7.987	7.986	7.986
Face4	125×125	7.662	7.257	7.139	7.987	7.986	7.987
baboo	512×512	7.707	7.474	7.752	7.998	7.998	7.998
n							
Peppe	512×512	7.339	7.496	7.058	7.998	7.998	7.998
rs							
4.1.01	256×256	6.420	6.446	6.381	7.996	7.996	7.996
4.1.04	256×256	7.255	7.270	6.783	7.995	7.996	7.996

calculation [42]. The formula of correlation coefficient ρ is:

$$\rho = \frac{E[(X - \mu_X)(Y - \mu_Y)]}{\sigma_X \sigma_Y}. \quad (19)$$

We calculated the plain image (baboon and pepper) as well as the corresponding cipher ones. The results are shown in Fig. 9. The correlation data for all images and comparison, including the aforementioned facial images, are recorded in Table IV and Table V.

D. Key Space and Sensitivity

In this paper, a total of five parameters are used, all set within the range of [0,10]. Three initial values are set within [0,1], and the type of variables is uniformly double. The system's key space can be calculated as $(10^{16})^5 \times (10^{16})^3 = 10^{128}$, which is larger than 2^{425} . Research indicates that cryptographic system must have a key space larger than

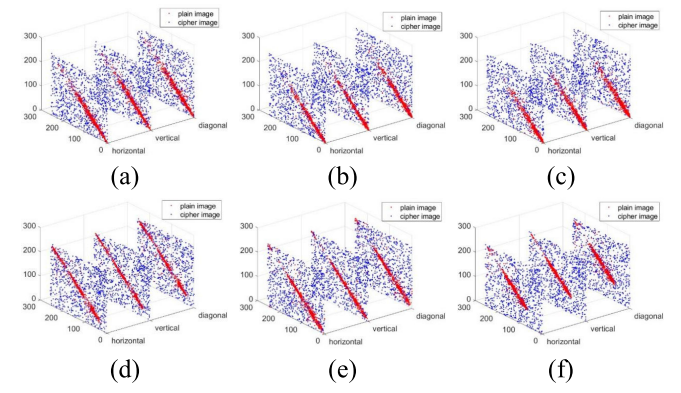


Fig. 9. Correlation diagram of:(a), (b), (c)4.1.01 and (d), (e), (f)4.1.04 from three channels.

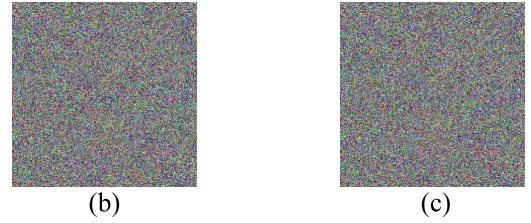
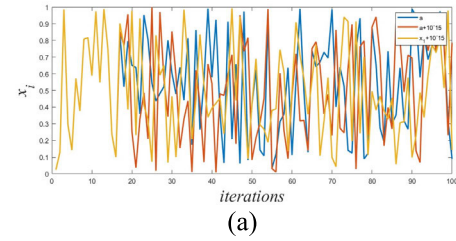


Fig. 10. Key sensitivity. (a)result of chaotic sequence when keys change; (b) decrypted result when a change; (c) decrypted result when x_1 change.

2^{256} to ensure it cannot be compromised by brute force attacks [44], and clearly, the encryption scheme in this paper meets this criterion. Additionally, we can further expand the key space by changing the parameter type from one double-precision floating-point to two double-precision floating-point numbers. Table VI shows the comparison of key space and entropy between different algorithm. Furthermore, since the PSO algorithm uses randomly selected plaintext images to generate keys, different plaintext images will produce different keys. Therefore, the encryption scheme in this paper can also effectively resist chosen-plaintext attacks.

A good cryptographic system should also be highly sensitive to key values [50]. Taking an image as an example, based on previous encryption and decryption results, we attempted to add an offset of 10^{-15} to the correct system parameter a and the initial value x_1 during decryption. The final decryption results are shown in Fig. 10.

E. Differential Attacks

NPCR (Number of Pixels Change Rate) and UACI (Unified Average Changing Intensity) are two crucial metrics used in differential cryptanalysis. These metrics are primarily used to measure the responsiveness of the encrypted image to minor

TABLE IV
CORRELATION BETWEEN ADJACENT PIXELS OF TESTED IMAGES

Image	Horizontal (10^{-3})			Vertical (10^{-3})			Diagonal (10^{-3})		
	Red	Green	Blue	Red	Green	Blue	Red	Green	Blue
Face1	6.1122	11.057	-9.0066	7.1802	-12.737	-8.65	-13.014	-3.0022	-0.2199
Face2	-7.6022	-5.3961	-2.9825	4.0073	2.3729	3.0506	-0.1770	4.1008	2.9345
Face3	-5.6467	-2.8107	-0.6144	-5.2439	-13.538	-0.0319	-3.2557	-2.5716	-12.322
Face4	-7.0431	-8.9211	7.6292	6.3436	1.2404	-8.7144	-3.8613	5.6476	-5.8086
baboon	-0.9464	0.8934	-2.3211	1.5403	1.4408	1.5048	0.9071	-1.0816	-0.9687
Peppers	2.1948	-1.9502	0.4686	-1.3982	1.5323	-0.0386	-0.9032	0.3481	0.3018
4.1.01	6.3594	-2.3142	3.4930	2.7229	4.1650	2.4451	-0.7051	-2.4149	-0.0404
4.1.04	10.975	-3.6882	1.916	-4.0755	2.2114	-0.6461	2.2438	-3.4545	-3.7864

TABLE V
COMPARISON OF MEAN CORRELATION BETWEEN ADJACENT PIXELS

Algorithm	Direction(10^{-3})		
	Horizontal	Vertical	Diagonal
Ref.[20]	0.8490	0.6800	0.2760
Ref.[27]	-0.1022	0.3399	0.2489
Ref.[46]	1.253	0.0896	0.0074
Ref.[47]	-4.9	6.7	0.6
Ref.[48]	-1.5	2.3	2.1
Ref.[49]	3.650	0.8233	3.360
Ours	-0.3514	-0.5548	0.9452

TABLE VI
COMPARISON OF KEY SPACE AND ENTROPY

Algorithm	Key space	Entropy of encrypted images			
		Red	Green	Blue	average
Ref.[20]	2^{256}	7.9993	7.9994	7.9991	7.9992
Ref.[27]	2^{256}	/	/	/	7.9898
Ref.[46]	2^{425}	7.9912	7.9913	7.9914	7.9913
Ref.[47]	2^{512}	7.9939	7.9939	7.9939	7.9939
Ref.[48]	2^{512}	7.9023	7.9029	7.9023	7.9025
Ref.[49]	2^{399}	7.9987	7.9985	7.9984	7.9985
Ours	2^{425}	7.9968	7.9970	7.9968	7.9969

changes in the original image. The formulas for calculating NPCR and UACI are provided [51], as shown in (20) and (21).

$$\left\{ \begin{array}{l} NPCR = \frac{\sum_{i=1}^m \sum_{j=1}^n \sum_{k=1}^3 D(i, j, k)}{m \times n \times 3} \\ D(i, j, k) = \begin{cases} 1, & P_1(i, j, k) \neq P_2(i, j, k) \\ 0, & elsewhere \end{cases} \end{array} \right. \quad (20)$$

$$UACI = \frac{1}{m \times n \times 3} \left(\sum_{i=1}^m \sum_{j=1}^n \sum_{k=1}^3 \frac{|P_1(i, j, k) - P_2(i, j, k)|}{255} \right), \quad (21)$$

where (m, n) represents the dimensions of the image, P_1 and P_2 are the encryption results corresponding to plaintext images that differ by one bit in pixel value.

TABLE VII
CRITICAL VALUES OF THE NPCR AND UACI

Size	120×1	125×1	133×1	140×1	256×2	512×5
	20	25	33	40	56	12
$NPCR^-$	99.523	99.527	99.532	99.536	99.569	99.589
(%)	9	3	2	1	3	3
$UACI^-$	33.077	33.092	33.114	33.132	33.282	33.373
(%)	1	5	8	3	4	0
$UACI^+$	33.850	33.834	33.812	33.794	33.644	33.554
(%)	0	6	3	8	7	1

The ideal values for NPCR and UACI, when only one bit of the image is changed, are approximately 99.6094% and 33.4635%. The closer the calculated results are to the ideal value, the better. Depending on the significance level α and the size of the image, the critical values for NPCR and UACI also differ. The critical value of the NPCR can be calculated as (22).

$$NPCR_{\alpha}^- = \frac{L - \Phi^{-1}(\alpha) \sqrt{L/mn}}{L - 1}, \quad (22)$$

where L is the grey level, α is the significance level. Similarly, we can also obtain two critical values for UACI ($UACI^-$ and $UACI^+$) according to (23)-(25) [52].

$$\left\{ \begin{array}{l} UACI_{\alpha}^- = \mu_u - \Phi^{-1}(\alpha/2) \times \sigma_u \\ UACI_{\alpha}^+ = \mu_u + \Phi^{-1}(\alpha/2) \times \sigma_u \end{array} \right. \quad (23)$$

$$\mu_u = \frac{L + 2}{3L + 3}, \quad (24)$$

$$\sigma_u^2 = \frac{(L + 2)(L^2 + 2L + 3)}{18mnL(L + 1)^2}. \quad (25)$$

We set α to 0.05; the critical value of different size is calculated in Table VII. Table VIII and Table IX show the NPCR and UACI results of different input and different algorithm. Any results within this range are considered to pass the test, and it can be seen that all eight subjects tested in this paper passed.

F. Robustness Analysis

Encrypted images often receive interference from channel noise and attackers during transmission. A system with good

TABLE VIII
NPCR AND UACI OF TESTED IMAGES

Image	NPCR (%)			UACI (%)			result
	Red	Green	Blue	Red	Green	Blue	
Face1	99.61	99.60	99.60	33.51	33.45	33.54	Pass
	35	52	69	53	40	66	
Face2	99.60	99.60	99.60	33.44	33.43	33.44	Pass
	52	51	51	91	89	41	
Face3	99.60	99.61	99.60	33.41	33.46	33.44	Pass
	78	50	80	99	83	91	
Face4	99.60	99.60	99.61	33.48	33.44	33.49	Pass
	38	95	64	03	67	63	
baboon	99.61	99.61	99.60	33.48	33.45	33.44	Pass
	29	43	59	55	85	29	
Pepers	99.61	99.60	99.61	33.49	33.44	33.44	Pass
	00	82	11	23	70	17	
4.1.01	99.61	99.61	99.61	33.41	33.43	33.54	Pass
	32	61	41	76	58	24	
4.1.04	99.61	99.61	99.61	33.43	33.44	33.42	Pass
	62	70	09	88	07	35	

TABLE IX
NPCR AND UACI BETWEEN DIFFERENT ALGORITHM

algorithm	NPCR(%)	UACI(%)
Ref. [20]	99.6075	33.4615
Ref. [27]	99.6083	33.4657
Ref. [46]	99.6183	33.4783
Ref. [47]	99.6100	33.4629
Ref. [48]	99.6101	33.4751
Ref. [49]	99.6101	33.8414
Ours	99.6105	33.4615

TABLE X
SPEED TEST FOR PROPOSED ALGORITHM

image	size	Face size	Global(s)	Face only(s)
Face1	286*400	133*133	0.3440	0.1356
Face2	261*400	140*140	0.3236	0.1438
Face3	273*400	120*120	0.3381	0.1264
Face4	296*400	125*125	0.3100	0.1264
Peppers	512*512	/	0.7411	/
Baboon	512*512	/	0.7320	/
4.1.01	256*256	/	0.2355	/
4.1.04	256*256	/	0.2373	/

robustness can withstand certain levels of external attacks, and the encrypted images can still reveal some plaintext information after being decrypted.

In this section, we introduced noise into the encrypted images and decrypted them. We tested the robustness of the encryption system using two common types of noise (salt and pepper noise and Gaussian noise) and cropping attacks. Fig. 11. displays the results of these tests, showing that the encryption system discussed in this paper has good robustness.

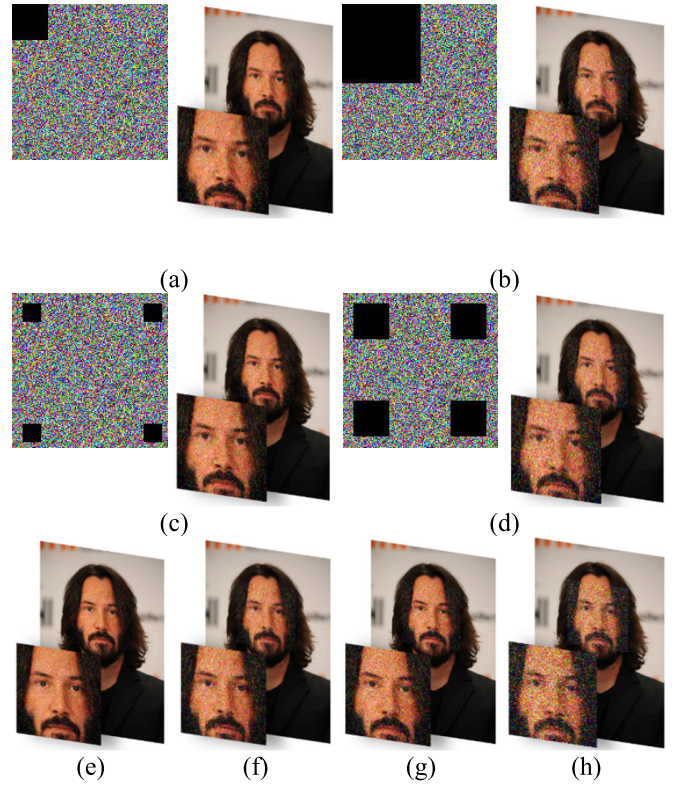


Fig. 11. Decryption results under different attacks.(a),(c)1/16 cropping attack; (b),(d)1/4 cropping attack; (e)0.5% Gaussian noise; (f) 2% Gaussian noise; (g) 5% salt-and-pepper noise; (h) 15% salt-and-pepper noise.

TABLE XI
ENCRYPTION TIME OF DIFFERENT ALGORITHMS

algorithm	Ref. [20]	Ref. [27]	Ref. [46]	Ref. [47]	Ref. [49]	ours
Time(s)	1.3053	/	1.8632	1.4310	/	0.6629
CC(10 ⁹)	2.8717	/	4.2854	4.1499	/	2.5853

G. Encryption Speed Test

We first analyzed the encryption times for various algorithms, including the one proposed in this paper, for encrypting color images of size 512×512 . Each image was encrypted 12 times, with the highest and lowest values removed and the average taken. The final results are shown in Table X. Additionally, we analyzed the time comparison for global encryption versus selective encryption using the algorithm proposed in this paper. We used images 1 to 4 as inputs, encrypting both the facial regions and the entire images. Considering the impact of different experimental platforms, we introduce clock cycles to compare the computational complexity between different algorithms [53]. The calculation formula is as follows:

$$CC = t \times \text{Processor Clock Frequency(Hz)} \quad (26)$$

Here, CC represents Clock Cycles, and t represents the time required for the algorithm to execute, in seconds. The final results are shown in Table XI. The clock cycles required by our algorithm are fewer compared to other literature.

TABLE XII
SPEED TEST FOR PROPOSED ALGORITHM

Sub-tests	Ref. [54]		Ref. [55]		ours	
	P-value	Proportion	P-value	Proportion	P-value	Proportion
Frequency (Monobit)	0.972	99/100	0.543	10/10	0.679	99/100
Frequency (block)	0.798	98/100	0.058	10/10	0.225	99/100
Runs	0.699	97/100	0.543	10/10	0.514	100/100
Longest Run	0.534	98/100	0.134	10/10	0.091	99/100
Binary Matrix Rank	0.202	99/100	0.993	10/10	0.081	98/100
FFT	0.401	98/100	0.036	10/10	0.419	100/100
Non-Overlapping Template	0.998	100/100	0.749	10/10	0.898	100/100
Overlapping Template	0.304	98/100	0.362	10/10	0.720	98/100
Maurer Universal Statistic	0.740	100/100	/	/	0.534	99/100
Linear Complexity	0.514	99/100	0.748	9/10	0.798	99/100
Serial(1)	0.760	100/100	0.029	10/10	0.494	99/100
Serial(2)	/	/	0.535	10/10	0.779	100/100
Approximate Entropy	0.911	99/100	0.524	9/10	0.401	99/100
Cumulative Sums (Forward)	0.964	98/100	0.542	10/10	0.091	98/100
Cumulative Sums (Reverse)	/	/	0.542	10/10	0.978	99/100
Random Excursions	0.881	100/100	/	/	0.804	100/100
Random Excursions Variant	0.952	100/100	/	/	0.996	100/100

H. NIST Statistical Test

The NIST (National Institute of Standards and Technology) test suite is a collection of statistical tests developed by the NIST to assess the randomness of digital sequences. It is primarily used to evaluate the randomness of sequences generated by cryptographic algorithms, random number generators, and similar applications.

NIST test can confirm whether pseudo-random number sequences indeed possess a high level of randomness, which is crucial for ensuring the security of encryption algorithms [54], [55]. Using the default parameters, the sequence x_n is iterated. 100 sets of data are randomly selected, each of length 1 million. We conducted NIST tests on these sets, and the passed numbers and p-values of the final results are shown in Table XII. The chaotic sequences generated passed all 17 tests,

and compared to other systems, they exhibited higher overall p-values.

VI. CONCLUSION

This paper proposes a new 3D chaotic system and designs a new selective encryption scheme for color images. Firstly, we use a PSO algorithm associated with the plaintext image to search for the optimal key. Then, we use DeepFace to achieve facial region recognition and extraction. The encryption of facial images is divided into two steps: a row-column alternation cyclic shift of the image pixel positions and the STP diffusion. Images' information entropy, histogram, correlation and SSIM show that the encrypted image exhibits a random distribution in space. Meanwhile, the encryption system exhibit resistance to different attacks. Also, our scheme achieves a shorter time consumption when encrypting the same amount of information. In future work, we plan to further modify the optimization algorithm to improve its optimization iteration efficiency, redesign the algorithm, and apply it to video encryption. Finally, we hope to implement the lightweight deployment of our overall encryption process and its application in secure transmission using the FPGA platform.

REFERENCES

- [1] N.-R. Zhou, L.-L. Hu, Z.-W. Huang, M.-M. Wang, and G.-S. Luo, "Novel multiple color images encryption and decryption scheme based on a bit-level extension algorithm," *Expert Syst. Appl.*, vol. 238, Mar. 2024, Art. no. 122052.
- [2] Q. Li et al., "Concealed attack for robust watermarking based on generative model and perceptual loss," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 8, pp. 5695–5706, Aug. 2022.
- [3] H. Zhang, H. Hu, and W. Ding, "VSDHS-CIEA: Color image encryption algorithm based on novel variable-structure discrete hyperchaotic system and cross-plane confusion strategy," *Inf. Sci.*, vol. 665, Apr. 2024, Art. no. 120332.
- [4] Y. Xian, X. Wang, and L. Teng, "Double parameters fractal sorting matrix and its application in image encryption," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 6, pp. 4028–4037, Jun. 2022.
- [5] L. Teng, X. Wang, and Y. Xian, "Image encryption algorithm based on a 2D-CLSS hyperchaotic map using simultaneous permutation and diffusion," *Inf. Sci.*, vol. 605, pp. 71–85, Aug. 2022.
- [6] M. Kumar, A. Iqbal, and P. Kumar, "A new RGB image encryption algorithm based on DNA encoding and elliptic curve Diffie–Hellman cryptography," *Signal Process.*, vol. 125, pp. 187–202, Aug. 2016.
- [7] X. Kang and R. Tao, "Color image encryption using pixel scrambling operator and reality-preserving MPFRHT," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 29, no. 7, pp. 1919–1932, Jul. 2019.
- [8] A. Ghaffari, "Image encryption-compression method via encryption based sparse decomposition," *Multimedia Tools Appl.*, vol. 83, no. 7, pp. 19129–19160, Jul. 2023.
- [9] Y. Li, C. Li, Y. Li, I. Moroz, and Y. Yang, "A joint image encryption based on a memristive rulkov neuron with controllable multistability and compressive sensing," *Chaos, Solitons Fractals*, vol. 182, May 2024, Art. no. 114800.
- [10] H. Gan, X. Wang, L. He, and J. Liu, "Learned two-step iterative shrinkage thresholding algorithm for deep compressive sensing," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 5, pp. 3943–3956, May 2024.
- [11] K. Yu, L. Chen, Z. Fu, Y. Wang, and T. Lu, "A coding layer robust reversible watermarking algorithm for digital image in multi-antenna system," *Signal Process.*, vol. 199, Oct. 2022, Art. no. 108630.
- [12] S. Geetha, P. Punithavathi, A. M. Infanteena, and S. S. S. Sindhu, "A literature review on image encryption techniques," *Int. J. Inf. Secur. Privacy*, vol. 12, no. 3, pp. 42–83, Jul. 2018.

- [13] M. Singh, N. Baranwal, K. N. Singh, A. K. Singh, and H. Zhou, "Deep learning-based biometric image feature extraction for securing medical images through data hiding and joint encryption-compression," *J. Inf. Secur. Appl.*, vol. 79, Dec. 2023, Art. no. 103628.
- [14] T. Gao and Z. Chen, "A new image encryption algorithm based on hyper-chaos," *Phys. Lett. A*, vol. 372, no. 4, pp. 394–400, Jan. 2008.
- [15] Y. Dong, G. Zhao, Y. Ma, Z. Pan, and R. Wu, "A novel image encryption scheme based on pseudo-random coupled map lattices with hybrid elementary cellular automata," *Inf. Sci.*, vol. 593, pp. 121–154, May 2022.
- [16] D. Ding et al., "Controllable multistability of fractional-order memristive coupled chaotic map and its application in medical image encryption," *Eur. Phys. J. Plus*, vol. 138, no. 10, p. 908, Oct. 2023.
- [17] M. Kumar, A. Saxena, and S. S. Vuppala, "A survey on chaos based image encryption techniques," in *Multimedia Security Using Chaotic Maps: Principles and Methodologies*, J. Smith, Ed., New York, NY, USA: Springer, Feb. 2020, pp. 1–26.
- [18] D. Ding et al., "Coexisting multi-stability of Hopfield neural network based on coupled fractional-order locally active memristor and its application in image encryption," *Nonlinear Dyn.*, vol. 108, no. 4, pp. 4433–4458, 2022.
- [19] H. Zhu, Y. Zhao, and Y. Song, "2D logistic-modulated-sine-coupling-logistic chaotic map for image encryption," *IEEE Access*, vol. 7, pp. 14081–14098, 2019.
- [20] H. Qiu, X. Xu, Z. Jiang, K. Sun, and C. Xiao, "A color image encryption algorithm based on hyperchaotic map and Rubik's cube scrambling," *Nonlinear Dyn.*, vol. 110, no. 3, pp. 2869–2887, Nov. 2022.
- [21] O. Kocak, U. Erkan, A. Toktas, and S. Gao, "PSO-based image encryption scheme using modular integrated logistic exponential map," *Expert Syst. Appl.*, vol. 237, Mar. 2024, Art. no. 121452.
- [22] F. Chen, Y. Yuan, H. He, M. Tian, and H.-M. Tai, "Multi-MSB compression based reversible data hiding scheme in encrypted images," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 3, pp. 905–916, Mar. 2021.
- [23] A. Y. Darani, Y. K. Yengejeh, H. Pakmanesh, and G. Navarro, "Image encryption algorithm based on a new 3D chaotic system using cellular automata," *Chaos, Solitons Fractals*, vol. 179, Feb. 2024, Art. no. 114396.
- [24] D. Ding et al., "An n-dimensional modulo chaotic system with expected Lyapunov exponents and its application in image encryption," *Chaos, Solitons Fractals*, vol. 174, Sep. 2023, Art. no. 113841.
- [25] R. Zhou and S. Yu, "Break an enhanced plaintext-related chaotic image encryption algorithm," *Chaos, Solitons Fractals*, vol. 181, Apr. 2024, Art. no. 114623.
- [26] S. Gao et al., "A 3D model encryption scheme based on a cascaded chaotic system," *Signal Process.*, vol. 202, Jan. 2023, Art. no. 108745.
- [27] S. Gao, R. Wu, X. Wang, J. Liu, Q. Li, and X. Tang, "EFR-CSTP: Encryption for face recognition based on the chaos and semi-tensor product theory," *Inf. Sci.*, vol. 621, pp. 766–781, Apr. 2023.
- [28] M. Lyle, P. Sarosh, and S. A. Parah, "Selective medical image encryption based on 3D Lorenz and logistic system," *Multimedia Tools Appl.*, vol. 83, no. 15, pp. 45553–45574, Oct. 2023.
- [29] Y. Shen, C. Tang, M. Xu, and Z. Lei, "Optical selective encryption based on the FRFCM algorithm and face biometric for the medical image," *Opt. Laser Technol.*, vol. 138, Jun. 2021, Art. no. 106911.
- [30] S. S. Jamal, M. M. Hazzazi, M. F. Khan, Z. Bassfar, A. Aljaedi, and Z. U. Islam, "Region of interest-based medical image encryption technique based on chaotic S-boxes," *Expert Syst. Appl.*, vol. 238, Mar. 2024, Art. no. 122030.
- [31] K. M. Hosny, M. A. Zaki, H. M. Hamza, M. M. Fouda, and N. A. Lashin, "Privacy protection in surveillance videos using block scrambling-based encryption and DCNN-based face detection," *IEEE Access*, vol. 10, pp. 106750–106769, 2022.
- [32] L. Du, L. Teng, H. Liu, and H. Lu, "Multiple face images encryption based on a new non-adjacent dynamic coupled mapping lattice," *Expert Syst. Appl.*, vol. 238, Mar. 2024, Art. no. 121728.
- [33] M. S. H. Mukta, J. Ahmad, A. Zaman, and S. Islam, "Attention and meta-heuristic based general self-efficacy prediction model from multimodal social media dataset," *IEEE Access*, vol. 12, pp. 36853–36873, 2024.
- [34] M. Wang and W. Deng, "Deep face recognition: A survey," *Neurocomputing*, vol. 429, pp. 215–244, Mar. 2021.
- [35] A. Shah, B. Ali, M. Habib, J. Frnda, I. Ullah, and M. S. Anwar, "An ensemble face recognition mechanism based on three-way decisions," *J. King Saud Univ.-Comput. Inf. Sci.*, vol. 35, no. 4, pp. 196–208, Apr. 2023.
- [36] X. Wang and Y. Li, "Chaotic image encryption algorithm based on hybrid multi-objective particle swarm optimization and DNA sequence," *Opt. Lasers Eng.*, vol. 137, Feb. 2021, Art. no. 106393.
- [37] L. S. Khan, M. M. Hazzazi, M. Khan, and S. S. Jamal, "A novel image encryption based on Rossler map diffusion and particle swarm optimization generated highly non-linear substitution boxes," *Chin. J. Phys.*, vol. 72, pp. 558–574, Aug. 2021.
- [38] S. Gao et al., "Asynchronous updating Boolean network encryption algorithm," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 8, pp. 4388–4400, Aug. 2023.
- [39] T. Gopalakrishnan and S. Ramakrishnan, "Image encryption using hyper-chaotic map for permutation and diffusion by multiple hyper-chaotic maps," *Wireless Pers. Commun.*, vol. 109, no. 1, pp. 437–454, Nov. 2019.
- [40] Z. Fan, X. Sun, J. Zhao, C. Zhang, and B. Du, "Dynamics analysis and feasibility verification of a 3D discrete memristive chaotic map with multi-vortex-like volume behavior," *Chaos, Solitons Fractals*, vol. 185, Aug. 2024, Art. no. 115070.
- [41] G.-Q. Si, H. Cao, and Y.-B. Zhang, "A new four-dimensional hyper-chaotic Lorenz system and its adaptive control," *Chin. Phys. B*, vol. 20, no. 1, Jan. 2011, Art. no. 010509.
- [42] M. Kumar, R. N. Mohapatra, S. Agarwal, G. Sathish, and S. N. Raw, "A new RGB image encryption using generalized vigenère-type table over symmetric group associated with virtual planet domain," *Multimedia Tools Appl.*, vol. 78, no. 8, pp. 10227–10263, Apr. 2019.
- [43] D. Ding, S. Chen, H. Zhang, Z. Yang, F. Jin, and X. Liu, "Firing pattern transition of fractional-order memristor-coupled Hindmarsh-Rose neurons model and its medical image encryption for region of interest," *Nonlinear Dyn.*, vol. 112, no. 12, pp. 10529–10554, Jun. 2024.
- [44] J. R. de Oliveira Neto, J. B. Lima, and D. Panario, "The design of a novel multiple-parameter fractional number-theoretic transform and its application to image encryption," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 8, pp. 2489–2502, Aug. 2020.
- [45] D. Ding, H. Zhu, H. Zhang, Z. Yang, and D. Xie, "An n-dimensional polynomial modulo chaotic map with controllable range of Lyapunov exponents and its application in color image encryption," *Chaos, Solitons Fractals*, vol. 185, Aug. 2024, Art. no. 115168.
- [46] L. Teng, X. Wang, F. Yang, and Y. Xian, "Color image encryption based on cross 2D hyperchaotic map using combined cycle shift scrambling and selecting diffusion," *Nonlinear Dyn.*, vol. 105, no. 2, pp. 1859–1876, 2021.
- [47] X. Wang and M. Zhao, "An image encryption algorithm based on hyperchaotic system and DNA coding," *Opt. Laser Technol.*, vol. 143, Nov. 2021, Art. no. 107316.
- [48] L. Li, "A novel chaotic map application in image encryption algorithm," *Expert Syst. Appl.*, vol. 252, Oct. 2024, Art. no. 124316.
- [49] M. Gong, X. Chai, Y. Lu, and Y. Zhang, "Exploiting four-dimensional chaotic systems with dissipation and optimized logical operations for secure image compression and encryption," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 8, pp. 7628–7642, Aug. 2024.
- [50] M. Kumar, P. Powduri, and A. Reddy, "An RGB image encryption using diffusion process associated with chaotic map," *J. Inf. Secur. Appl.*, vol. 21, pp. 20–30, Apr. 2015.
- [51] P. Liu, X. Wang, and Y. Su, "Image encryption via complementary embedding algorithm and new spatiotemporal chaotic system," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 5, pp. 2506–2519, May 2023.
- [52] Y. Zhang, "A new unified image encryption algorithm based on a lifting transformation and chaos," *Inf. Sci.*, vol. 547, pp. 307–327, Feb. 2021.
- [53] D. Clemente-López, J. M. Munoz-Pacheco, and J. D. J. Rangel-Magdaleno, "Experimental validation of IoT image encryption scheme based on a 5-D fractional hyperchaotic system and numba JIT compiler," *Internet Things*, vol. 25, Apr. 2024, Art. no. 101116.
- [54] Q. Cun, X. Tong, Z. Wang, and M. Zhang, "Selective image encryption method based on dynamic DNA coding and new chaotic map," *Optik*, vol. 243, Oct. 2021, Art. no. 167286.
- [55] M. Vijayakumar and A. Ahilan, "An optimized chaotic S-box for real-time image encryption scheme based on 4-dimensional memristive hyperchaotic map," *Ain Shams Eng. J.*, vol. 15, no. 4, Apr. 2024, Art. no. 102620.



Dawei Ding (Member, IEEE) received the B.S. and M.S. degrees from Anhui University, Hefei, China, in 1997 and 2004, respectively, and the Ph.D. degree from Shanghai Jiao Tong University, Shanghai, China, in 2008. From 2015 to 2016, he was a Visiting Scholar with Arizona State University. He is currently a Professor with the School of Electronic Information Engineering, Anhui University. His research interests include chaos, nonlinear circuit, memristor neural networks, image encryption, and secret communication.



Zongli Yang received the Ph.D. degree from Anhui University, Hefei, China. He is currently a Lecturer with the School of Electronic Information Engineering, Anhui University. His research interests include chaos, nonlinear circuit, secret communication, and image processing.



Dong Xie received the bachelor's degree in communication engineering from Anhui University, Hefei, China, where he is currently pursuing the master's degree with the School of Electronic Information Engineering. His research interests include chaotic map and image encryption.



Hongwei Zhang received the Ph.D. degree from Shanghai Jiao Tong University, Shanghai, China, in 2006. From 2012 to 2013, he was a Visiting Scholar with the City University of Hong Kong. He is currently an Associate Professor with the School of Electronic Information Engineering, Anhui University, Hefei, China. His research interests include mobile communication theory and application, cloud computing, big data analysis, the Internet of Things, and image processing. He is a member of Chinese Association for Artificial Intel-

ligence and a Senior Member of Anhui Communication Association.



Chu'an Liu received the bachelor's degree in communication engineering from Anhui University of Science and Technology, China, in 2024. He is currently pursuing the master's degree with the School of Electronic Information Engineering, Anhui university, Hefei. His research interests include chaotic map and image encryption.